

GPS parsing M9N GNSS receiver

Background information

GPS Data has information which needs to be transmitted to devices and people, the General standard is the NMEA 0183 which are human readable sentences used to communicate sensor data like position speed and heading, the understanding of these sentences are useful for the purpose of decoding and using GPS data in embedded systems.

This document serves as a guideline to decoding these messages to be used with microcontrollers or parsed using Python on a OS. The receiver to be used will be the M9N as example since it uses NMEA 0183 on default.

Decoding the sentence

\$XXXX,field1,field2,field3,...,fieldN*CC<CR><LF>

Part	Meaning
\$	Start of sentence
XX	Talker ID – which constellation/system is talking
YYY	Message type – what kind of data follows
,	Comma-delimited fields
*CC	Checksum (hex XOR of everything between \$ and *)
<CR><LF>	Carriage return + line feed (end of line)

Every sentence starts with a \$ sign, this is the main signal to the start of a message which can be used as a trigger to start the data acquisition per sentence. The second field is XX which is used to define which constellation is used for the current positioning. The third field gives the type of data being transmitted, this can be heading, velocity or position.

Talker ID's

ID	Constellation/System
GP	GPS (USA)
GL	GLONASS (Russia)
GA	Galileo (EU)
GB	BeiDou (China)
GN	Multi-GNSS (combined data from all constellations)

The Talker ID is used to identify which constellation is being used, theres 4 types which are used by the M9N GPS module which are GPS(USA), GLONASS (Russia), Galileo (EU) and Multi-GNSS which uses the data from all.

Message types

\$xxRMC

\$xxVTG

\$xxGGA

\$xxGSA

\$xxGSV

\$xxGGL

Min GNSS data (\$xxRMC)

Field #	Name	Your Sample	With Fix Example	Description
0	Sentence ID	\$GNRMC	\$GNRMC	Talker + message type
1	UTC Time	(empty)	112304.00	HHMMSS.ss – 11:23:04.00 UTC
2	Status	V	A	A = Active/valid, V = Invalid
3	Latitude	(empty)	5130.1234	ddmm.mmmm (degrees + decimal minutes)
4	N/S Indicator	(empty)	N	N = North, S = South
5	Longitude	(empty)	00215.5678	dddmm.mmmm
6	E/W Indicator	(empty)	W	E = East, W = West
7	Speed Over Ground	(empty)	12.5	Knots
8	Course Over Ground	(empty)	045.3	Degrees true
9	Date	(empty)	080926	DDMMYY – 8 Sep 2026
10	Magnetic Variation	(empty)	2.1	Degrees (empty if unknown)
11	Mag Var Direction	(empty)	W	E or W
12	Mode	N	A	N=No fix, A=Autonomous, D=DGPS, E=Estimated
13	Checksum	*37	*37	Hex XOR of chars between \$ and *

Track Made Good and GS (\$xxVTG)

Field #	Name	Your Sample	With Fix Example	Description
0	Sentence ID	\$GNVTG	\$GNVTG	Talker + message type
1	Course True	(empty)	045.3	Degrees relative to true north
2	Reference True	T	T	Fixed: T = True
3	Course Magnetic	(empty)	047.4	Degrees relative to magnetic north
4	Reference Mag	M	M	Fixed: M = Magnetic
5	Speed Knots	(empty)	12.5	Speed in nautical miles per hour
6	Unit Knots	N	N	Fixed: N = Knots
7	Speed KPH	(empty)	23.1	Speed in kilometers per hour
8	Unit KPH	K	K	Fixed: K = km/h
9	Mode	N	A	N=No fix, A=Autonomous, D=DGPS, E=Estimated
10	Checksum	*2E	*2E	Hex XOR checksum

Global Positioning system FIX data (\$xxGGA)

Field #	Name	Your Sample	With Fix Example	Description
0	Sentence ID	\$GNGGA	\$GNGGA	Talker + message type
1	UTC Time	(empty)	112304.00	HHMMSS.ss
2	Latitude	(empty)	5130.1234	ddmm.mmmm
3	N/S Indicator	(empty)	N	N or S
4	Longitude	(empty)	00215.5678	dddmm.mmmm
5	E/W Indicator	(empty)	W	E or W
6	Fix Quality	0	1	0=No fix, 1=GPS, 2=DGPS, 4=RTK fixed, 5=RTK float, 6=Dead reckoning
7	# Satellites	00	12	Number of satellites used (00–99)
8	HDOP	99.99	1.23	Horizontal Dilution of Precision – lower is better
9	Altitude MSL	(empty)	123.4	Meters above mean sea level
10	Altitude Unit	M	M	Fixed: M = Meters
11	Geoid Separation	(empty)	–45.6	Difference between WGS–84 ellipsoid and MSL (meters)
12	Geoid Unit	M	M	Fixed: M = Meters
13	Age of DGPS	(empty)	2.5	Seconds since last DGPS update (empty if not DGPS)
14	DGPS Station ID	(empty)	0123	Differential base station ID (0000–1023)
15	Checksum	*56	*56	Hex XOR checksum

GNSS DOP and Active Satellites (\$xxGSA)

Field #	Name	Your Sample	With Fix Example	Description
0	Sentence ID	\$GNGSA	\$GNGSA	Talker + message type
1	Mode	A	A	M = Manual 2D/3D forced, A = Automatic
2	Fix Type	1	3	1=No fix, 2=2D fix, 3=3D fix
3–14	Satellite PRNs	(all empty)	03,07,08,11,...	PRN (ID) numbers of up to 12 satellites used in solution
15	PDOP	99.99	1.45	Position Dilution of Precision
16	HDOP	99.99	1.23	Horizontal DOP
17	VDOP	99.99	0.78	Vertical DOP
18	System ID	1	1	1=GPS, 2=GLONASS, 3=Galileo, 4=BeiDou
19	Checksum	*33	*33	Hex XOR checksum

Satellites in view (\$xxGSV)

Field #	Name	Your Sample	With Fix Example	Description
0	Sentence ID	\$GPGSV	\$GPGSV	Talker + message type (one per constellation)
1	Total Messages	1	2	Total number of GSV sentences for this system
2	Message Number	1	1	This sentence number (1 of Total)
3	Satellites in View	00	07	Total satellites visible for this constellation
4	Sat 1 PRN	(empty)	03	Satellite PRN / ID number
5	Sat 1 Elevation	(empty)	45	Elevation angle in degrees (0-90)
6	Sat 1 Azimuth	(empty)	123	Azimuth angle in degrees (0-359)
7	Sat 1 SNR	(empty)	42	Signal-to-noise ratio in dB-Hz (null if not tracking)
8-11	Sat 2	(empty)	07,12,045,38	Same 4 fields for 2nd satellite
12-15	Sat 3	(empty)	08,67,210,44	Same 4 fields for 3rd satellite
16-19	Sat 4	(empty)	11,15,089,35	Same 4 fields for 4th satellite
20	System ID	1	1	1=GPS, 2=GLONASS, 3=Galileo, 4=BeiDou
21	Checksum	*64	*64	Hex XOR checksum

Geographic Position, Latitude/Longitude (\$xxGLL)

Field #	Name	Your Sample	With Fix Example	Description
0	Sentence ID	\$GNGLL	\$GNGLL	Talker + message type
1	Latitude	(empty)	5130.1234	ddmm.mmmm
2	N/S Indicator	(empty)	N	N or S
3	Longitude	(empty)	00215.5678	dddmm.mmmm
4	E/W Indicator	(empty)	W	E or W
5	UTC Time	(empty)	112304.00	HHMMSS.ss
6	Status	V	A	A = Valid, V = Invalid
7	Mode	N	A	N=No fix, A=Autonomous, D=DGPS, E=Estimated
8	Checksum	*7A	*7A	Hex XOR checksum

Quick fix vs No fix “cheat sheet”

Sentence	Check This Field	No Fix Value	Good Fix Value
RMC	Field 2 (Status)	V	A
GGA	Field 6 (Quality)	0	1 or higher
GGA	Field 7 (# Sats)	00	08–30+
GSA	Field 2 (Fix Type)	1	2 or 3
GSA	Fields 15–17 (DOP)	99.99	< 2.0
GSV	Field 3 (In View)	00	04–20+
GLL	Field 6 (Status)	V	A